

The Utility Problem

As a significant trend in recent years, the adaptation rate of the behind-the-meter microgeneration is getting larger and larger. There are multiple attractive reasons why electricity consumers install these distributed microgeneration sources like solar photovoltaics (PV) or micro wind turbines. Among these reasons, it is worth mentioning offsetting the energy costs, reducing the home's environmental impact or contributing to energy independence. On the distribution utility side, these microgeneration sources bring multiple benefits and opportunities. The utility benefits from microgeneration by decreasing the need for capacity upgrades in certain areas, reducing transmission congestion, or offering service area consumers backup in case of an outage - these can be often summarized as non-wire alternatives (NWA). However, it is important to mention that the microgeneration benefits for distribution utility are often materialized only when the utility knows that the microgeneration is installed behind the utility energy meter (behind-the-meter).

Unfortunately, this is not always the case. There might be multiple reasons why utilities are missing records about the behind-the-meter microgeneration. Firstly, there is a large influx of microgeneration into the utility service areas, and many energy consumers install the microgeneration sources without reporting it to the utility. Secondly, the information flow or record-keeping process in utilities is often not ideal, so the records are not making their way into the system of records. These situations are not just a lost opportunity for utility and consumers to participate in various microgeneration and distribution energy resources (DER) programs but also pose a potential safety risk for utility personnel.

Safety is a concern when work needs to be performed on the grid's infrastructure (like wires, conductors, insulators, transformers, etc.), and the utility crew (linemen) is dispatched to the area to conduct the required activity. Many of the grid's equipment operations require work where the grid voltage can't be present because it might result in electric shock - electrocution. Utility linemen have multiple tools and procedures to prevent the voltage from reaching the equipment which needs to be worked on. The most common technique is disconnection/insulation from the source of voltage. This usually means disconnection from the upstream direction of the grid (leading to the higher voltage level substation). However, more and more distributed microgeneration assets exist in service areas, so it is important to realize that these are also sources of voltage. For these reasons, linemen often consult the utility system of records for the presence of microgeneration to ensure that no other sources of voltage are connected to the targeted equipment. Nevertheless, the missing utility records may lead linemen to false assumptions, and safety accidents might potentially occur.

For these reasons, it is beneficial to be able to proactively identify situations when the utility system of records is missing behind-the-meter microgeneration and generation is unexpectedly present on the utility meter. Successfully conducting such an analysis should result in increased safety for utility crews, which is invaluable.

The Data Solutions and Methods

The identification method proposed in this use case document assumes that the utility is equipped with smart meters capable of recording usage data in generation and consumption quadrants. The identification is based on a comparison of behind-the-meter microgeneration records with smart meter generation quadrant registers. For an analyst to conduct this type of analysis, two main data sets are required. First is the Client Information System (CIS) data which is usually a system of records for information about behind-the-meter microgeneration. The second dataset is the billing/AMI time series from the utility meters.

With the Awesense Energy Transition Platform, the creation of such identification analysis is greatly simplified. Firstly, all required datasets are native to Awesense's Energy Data Model (EDM) schema, so all required information is stored in an orderly and easily accessible manner. Secondly, the EDM's APIs offers simplified functions and calls, which makes working with time series much easier.

To simply summarize this type of analysis which is built on Awesense EDM, we can include:

1. Identify all meters/consumers/accounts where the generation register contains non-null/non-zero values.
 - a. This can be achieved by calling EDM's data source function (*ts_data_source()*).
2. Compare this list with the behind-the-meter microgeneration record.
3. Flag all meters/consumers/accounts where the discrepancy exists so the unexpected generation is present.

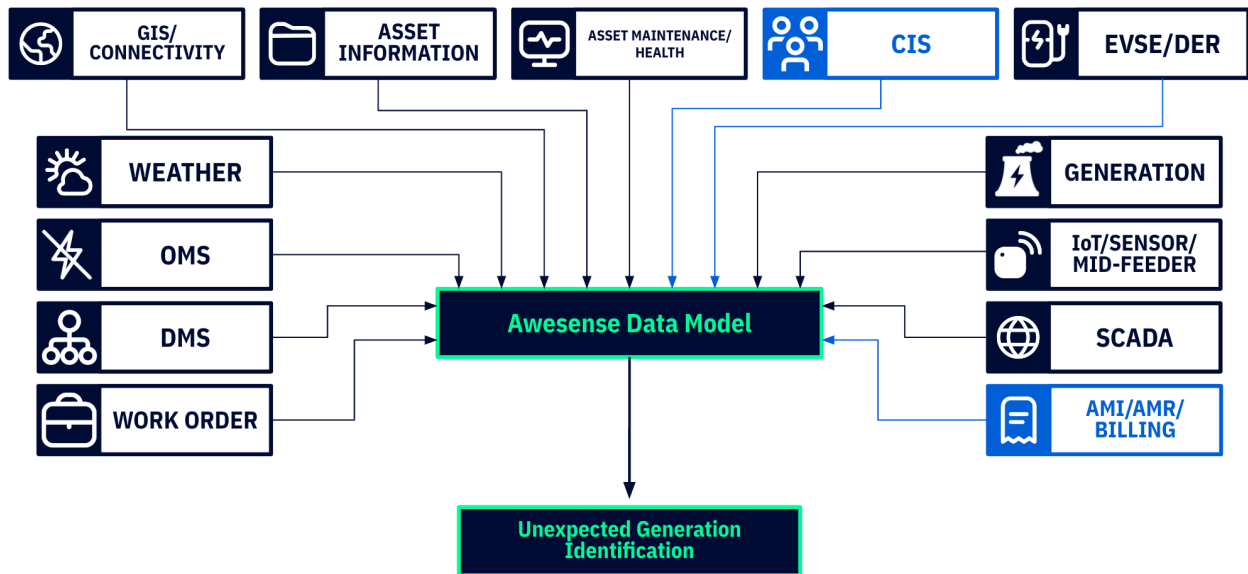
Unexpected Generation Identification

[UC20-01] - Awesense Energy Transition Platform Use Case



Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



Available Use Case Implementations:

Users of the Awesense Energy Transition Platform can access this use case by the following means:

1. GitHub link to the Jupyter notebook can be provided upon request.